20 n is an integer such that $3n + 2 \leqslant 14$ and $\dfrac{6n}{n^2 + 5} > 1$

Find all the possible values of n.

$3n + 2 \leq 14$

-2 -2

$3n \leq 12$

1 mark
Solve the linear inequality

$\div 3$ $\div 3$

$n \leq 4$

Since $n^2 + 5$ is positive, multiply without reversing the inequality:

$6n > n^2 + 5$, so $n^2 - 6n + 5 < 0$.

1 mark
Form a quadratic inequality

Product 5; sum −6: use −5 and −1.

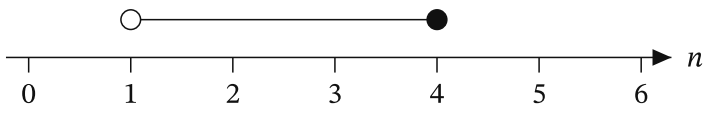
×	n	-5
n	n^2	$-5n$
-1	$-n$	5

$(n - 1)(n - 5) < 0$

1 mark
Factorise to find boundary roots

Test $n = 0$: $+5$; $n = 3$: -4; $n = 6$: $+5$.

Thus $1 < n < 5$. Combine this with $n \leq 4$:



1 mark
Keep $1 < n \leq 4$

n is an integer, so the possible values are 2, 3 and 4.

1 mark
List all allowed integers

2, 3, 4

(Total for Question 20 is 5 marks)